Oral Exam Syllabus

Avinash Iyer

October 2026

The oral exam for Avinash Iyer will be administered by Professors Ben Hayes and [either Mikhail Ershov or Thomas Koberda] on December 4, 2026 focused on Functional Analysis and Geometric Group Theory. The exam will consist of questions on both topics.

Functional Analysis Topics:

- Locally Convex Spaces, topics covering Chapter IV §1–§3 of Conway's *A Course in Functional Analysis*
  - Definitions and examples
  - Metrizability and Normability
  - Geometric Hahn–Banach
- Weak topologies and extremal structure, topics covering Chapter V §1–§10 §12–§13 of Conway's *A Course in Functional Analysis*
  - Dual spaces of locally convex topologies
  - Banach–Alaoglu Theorem
  - Reflexivity, Separability, and Metrizability
  - Stone–Čech compactification
  - Krein–Milman theorem
  - Schauder Fixed-Point Theorem, Ryll–Nardzewski Fixed-Point Theorem
  - Krein–Smulian Theorem
- C^*-algebras and von Neumann algebras, topics covering Chapter VIII §1–§3, §5 of Conway's *A Course in Functional Analysis* and Sections 4.6 and 4.7 of Pedersen's *Analysis Now*
  - Properties and examples
  - Gelfand Transform, Gelfand–Naimark Theorem, Continuous Functional Calculus
  - Ordering properties
  - GNS Construction
  - von Neumann double commutant theorem
  - σ-topologies and the predual of a von Neumann algebra
- Spectral theory for normal operators, topics covering Chapter IX §1–§8 of Conway's *A Course in Functional Analysis* and Sections 4.4–4.6 of Pedersen's *Analysis Now*
  - spectral measures and projections
  - Borel functional calculus

- the characterization of abelian von Neumann algebras